

Advanced Python Project (First semester (2025/2026))

Supermarket Sales Analysis (20 Marks) Deadline: 10/1/2026

Instructor Name: Dr. Ali Azwai and Dr. Ala Abuthawabeh

Objective

Students will use **Pandas**, **NumPy**, **Seaborn**, and **Matplotlib** to perform deep data analysis on the **Supermarket Sales Dataset**.

They will clean, explore, visualize, and interpret real-world sales data.

Supermarket Sales Dataset

Dataset: Supermarket Sales – Kaggle (Public)

Link: <https://www.kaggle.com/datasets/aungpyaeap/supermarket-sales>

Description (Students)

In this assignment, you will work with the **Supermarket Sales Dataset**, which contains real transaction records for supermarkets in Myanmar. You will use Python (Pandas, NumPy, Matplotlib, Seaborn) to perform data exploration, cleaning, visualization, and statistical analysis.

Your task is to complete **all steps below**, answer the required questions, and submit both:

1. A **Jupyter Notebook** or A **Colad (file)** with code and graphs
2. A **PDF report** summarizing your findings

Grading Breakdown (20 Marks)

Section	Task	Marks
1. Load & Inspect Data	Import dataset, show first rows, info, describe	2 marks
2. Data Cleaning	Handle missing values, adjust datatypes	2 marks
3. Exploratory Analysis	Branch stats, customer types, gender, payment	2 marks
4. Sales & Revenue Analysis	Revenue metrics, product line stats, correlations	2 marks
5. Visualization	6 professional visualizations	4 marks
6. Advanced Questions	Answer 5 analytical questions	4 marks

Q1: Which branch generates the highest revenue? Why might that be?

Q2: Do members spend more than normal customers?

Q3: Which payment method has the highest usage? Why?

Q4: Which product line has the highest average rating?

Q5: Is there a relationship between unit price and quantity purchased?

7. Submit code on elearning	send Github url	1 marks
8. Discussion	Run and presentation (record Video in English)	3 marks